Allocation Score Manuscript

Draft 2

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Abstract

In the development and assessment of infectious disease forecasts, modelers stand to benefit from evaluation measures that express a forecast's success in settings of public health consequence. The integrated allocation score (IAS) is a scoring rule that measures the units of unaddressed resource need in an idealized hypothetical scenario where a forecast directly informs allocation decisions. COVID-19 weekly incident case forecasts for 57 counties in California were compiled from the COVID-19 Forecast Hub and assessed by IAS over 13 weeks of the COVID-19 omicron wave at the 1- and 3-week horizons, using resource constraints informed by ground truth case counts. At each target date, an epidemiologically-naive resource allocation based on county populations was scored as a benchmark comparison, as well as an allocation based on the most recent case counts. Model rankings by IAS were found to align broadly with rankings by mean weighted interval score (MWIS), a commonly used scoring rule for probabilistic forecasts. Comparison of model IAS values to the control, however, found that forecast-informed allocations frequently failed to outperform the population-based allocation, particularly at the highest-burden weeks in the epidemic, suggesting that multi-week ahead case forecasts during the largest COVID-19 surge were not up to the task of informing resource allocation decision-making. This work underscores the importance of exploring whether model optimization targeted for specific metrics could improve forecast accuracy for specific decision-making. It also highlights the need for additional decision-specific metrics to be identified and developed to improve the utility of forecasts in public health practice.

Introduction

Forecast Use in Decision-Making and Need for Use-Aware Evaluation Metrics

In recent years, infectious disease forecasting has been used increasingly in public health surveillance and outbreak response. Accurate, timely forecasts of infectious disease indicators provide state, tribal, local, and territorial (STLT) public health practitioners with advance notice of local and national trends, enabling for rapid response to disease burden spikes, both through pharmaceutical and nonpharmaceutical interventions and through public information-sharing. Well-performing forecasts have the potential to contribute substantially to existing arsenals of public health tools, increasing the effectiveness of agencies' health-promotion efforts. Most notably, infectious disease forecasts came into widespread usage during the COVID-19 pandemic, through projects such as the COVID-19 Forecast Hub("Home - COVID 19 Forecast Hub," n.d.). Additional large scale examples of forecast adoption include CDC FluSight(CDC 2026) and the Dengue Forecasting Project(Johansson et al. 2019), in which forecasts submitted by modeling groups

were compiled, assessed, and used to inform both local and national public health policy. Modeling groups themselves recommend use of their forecasts for a range of objectives, including public communications, advance notice of spikes for hospitals and health departments, and to aid decisions around the allocation of medical resources and staffing.(Lutz et al. 2019) Indeed, a recent review of COVID-19 forecast usage found that, during the COVID-19 pandemic, jurisdictions applied these forecasts to uses such as understanding transmission dynamics, directing planning, and guiding resource management and allocation. (Rao et al. 2025)

However, as jurisdictions increasingly adopt these tools, a number of issues remain, and of these, the potential for misalignment between what the models are good at predicting and the priorities of STLT practitioners is of particular relevance.[**CITE**] Misalignment risks the development of models that fail to answer relevant public health questions, or that prove unsuitable for use in high-stakes policy decision-making due to incorrectly defined forecasting targets, unreliable predictions, or lack of interpretability.(Metcalf, Edmunds, and Lessler 2015) In the case of forecasting for situational awareness, current forecasts are largely thought to capture the key features needed to understand and track epidemic trends. In contrast, these forecasts are less successful when judged for their ability to predict inflection points/peaks in epidemics or guide resource distribution across locations. For forecast implementation to be supportive in decision-making contexts, it is therefore necessary that models be created, evaluated, and adopted with both the public health need and the priorities of the acting public health agency in mind.

Forecast evaluation metrics that align closely with public health goals are a vital part of this alignment. In order to ensure that models perform well in practice, forecast scoring metrics are used to evaluate and rank forecasts, based on comparisons between predicted outcomes and ground truth results. The scoring metrics are typically calculated at multiple timepoints or time horizons and then averaged to ensure reliable assessments and comparisons between competing models. A range of scoring metrics exist to conduct these evaluations, each with differing assumptions and objects of measurement.(Bjerregård, Møller, and Madsen 2021) Any definition of a "best" forecast is dependent upon which qualities the metric values most – this may include the closeness of point estimates to the ground truth, the width of prediction intervals, or the correct relative ranking of predicted values across a group of targets.[**CITE CITE CITE**] For this reason, forecast evaluation should include metrics that approximate user priorities, so that the best-scoring models in the testing phase are indeed those the most suited to the forecasting use of interest. [**This is a super important point, and I wonder if we could dwell on it here for a little longer. Like, how do current metrics value different forecast features? Maybe give a few specific examples of metrics and what they value? (This would sort of be a more specific version of the post-em-dash phrase in the first sentence.)**]

In this case study, we present an example of the application of the integrated allocation score (IAS), an evaluation metric that scores infectious disease forecasts based on effective resource allocation across multiple forecast targets, in the context of the COVID-19 omicron wave in California. We hope that this study can serve as an example of IAS's usage in a highly uncertain public health context, and reveal its qualities through contrast with other scoring methods.

Allocation Score and Integrated Allocation Score

High accuracy and precision in forecasts of individual targets (e.g., case counts for a single county) may not necessarily translate to optimal decision-making in dynamic scenarios with multiple allocation targets (e.g., all the counties in a state). The distribution of resources under constraint is a high priority for public health policy-makers and decision-makers in epidemic contexts, such as in scenarios of personal protective equipment (PPE), vaccine, or ventilator distribution[**CITE**]. The allocation score (AS), introduced by Gerding et al.(Gerding et al. 2024), scores forecasts on their success in allocating limited resources across multiple locations.

For a given target date, this method interprets a forecast's predictions as location-specific estimates of resource need and uses these predictions to guide the allocation of a fixed quantity of resource. The forecast is then scored by comparing this forecast-informed distribution scheme to an "oracle distribution" produced from the the ground truth (ie. the true resource need). For both the oracle-known need and the forecasted need, the total resource underallocation across all locations is tallied, and the overall difference across all locations between the resource underallocation of the oracle distribution (considered to be unavoidable) and that of the forecast-informed distribution is the allocation score. Lower scores reflect less avoidable underallocation of resources by the forecast and therefore better predictions. This aligns with other commonly used forecast scoring metrics, such as WIS, in that a lower score is indicative of better performance.(Bracher et al. 2021)

To address uncertainty with regards to resource constraints, the integrated allocation score (IAS) is preferred over AS alone. IAS is determined by calculating AS across a series of possible resource constraints and producing an average, which has the option of being weighted based on the perceived probabilities of the possible resource constraints, or weighted equally across all plausible values.

In utilizing IAS, it is important to be aware of a few important attributes and their impacts on the score. First, it should be noted that the AS and IAS methods measure underallocation, not overallocation, in the distribution of resources to the locations, meaning that they do not penalize resource waste. Second, the measurement of total underallocation as sum of all targets means that locations with larger populations and/or resource needs will contribute more to the score than smaller ones, an assumption that is valid in field situations, but potentially disadvantages forecasts with strong performance in smaller locations. Furthermore, by measuring resources on the absolute scale, the scores do not evaluate misallocated resources as a percentage of a location's true need, meaning that the unit counts of underallocations have the same impact the score, regardless of how large they are relative to the true resource need. Third, the raw calculation of AS does not account for the predictive distribution of the forecasts being evaluated (see Methods), and so sharper predictions are not directly rewarded, though predictive distributions that are both sharp and close to the ground truth can be expected to yield better scores in practice. Finally, in this scheme, the correct prediction of absolute values is secondary to the capture of relative ranking, meaning that a forecast which predicts every location as having exactly double the ground truth need would score better on the AS than one where all but a one location was predicted perfectly.(Gerding et al. 2024)

Example: Allocation Score of the 1-week COVIDhub-baseline forecast for January 1st, 2022

As a illustrative example, we will visualize the calculation of an allocation score for a baseline model that makes flatline predictions based on the most recently observed count data. For the purpose of scoring, predictions of COVID-19 incident cases are used here as proxies for resource need and the resource is kept abstract, but assumed to be apportionable. For this example, we set a resource constraint equal to 90% of the total incident cases for the date of interest: for the week ending 01/01/2022, a total of 131,492 cases were reported, and so the resource constraint for this target date is set at 118,343 units of resource, 90% of this. These resources are distributed based on the county-level incident case predictions of the COVIDhub-baseline forecast for this target date and horizon. (See Methods for more details on the distribution of resources from a forecast in quantile format.) In Figure 1, we see a breakdown of this resource allocation, in which red shading indicates underallocation to a location, blue shading indicates overallocation, and purple shading indicates addressed need.

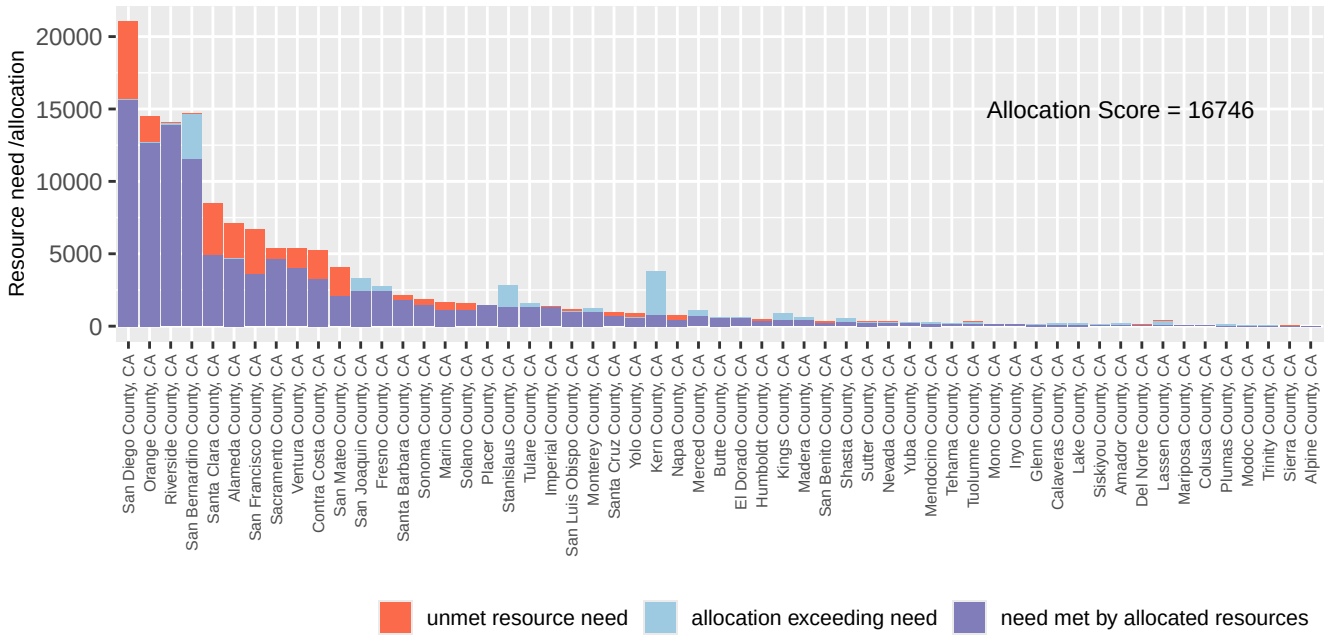


Figure 1: Visualization of an allocation score at for a model of flatline predictions based on recently observed count data, predicting case counts in California counties (with the exception of Los Angeles) for January 1st, 2022 at the 1 week horizon. The forecast predictions are used to inform the allocation of an abstract resource, set at a contraint of 90% of the ground truth case count for the date of interest. This is compared to an 'oracle' allocation informed by ground truth case counts. In the stacked bar charts, red shading indicates underallocation to a location, blue shading indicates overallocation, and purple shading indicates addressed need. When summed, avoidable underallocation (i.e. underallocation that is not inevitable due to lower-than-needed resources) is found to be 4,175 units.

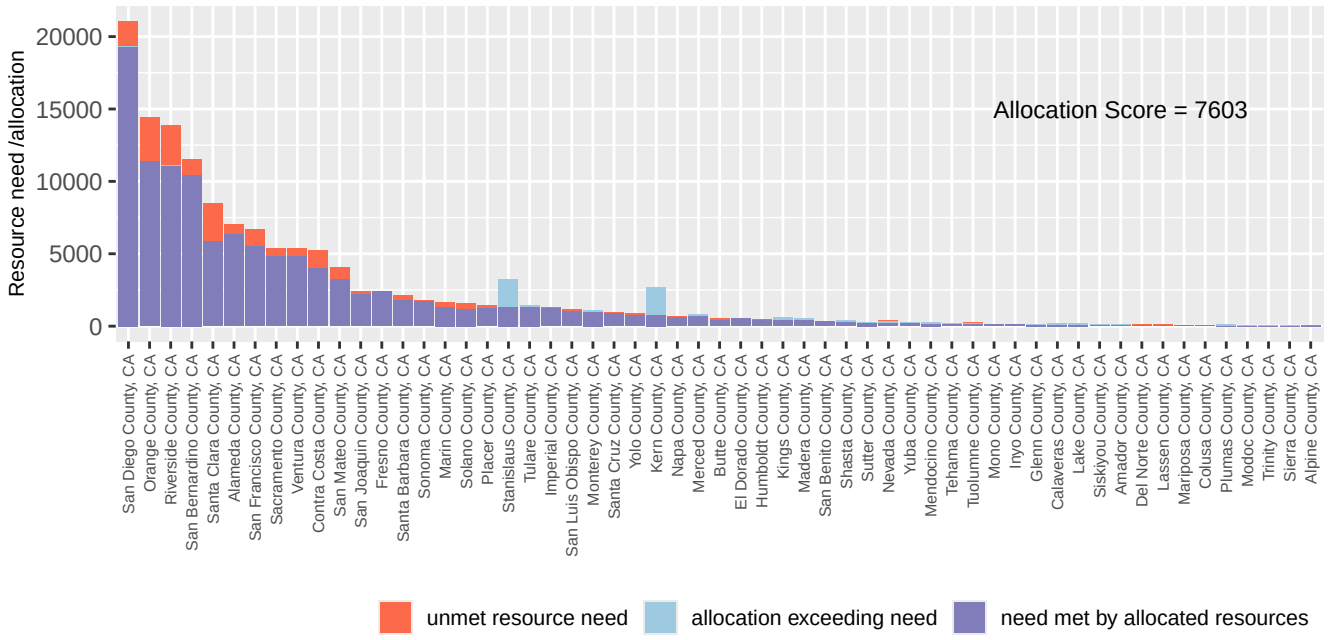


Figure 2: Visualization of an allocation score at for a model of flatline predictions based on recently observed count data, predicting case counts in California counties (with the exception of Los Angeles) for January 1st, 2022 at the 1 week horizon. The forecast predictions are used to inform the allocation of an abstract resource, set at a contraint of 90% of the ground truth case count for the date of interest. This is compared to an 'oracle' allocation informed by ground truth case counts. In the stacked bar charts, red shading indicates underallocation to a location, blue shading indicates overallocation, and purple shading indicates addressed need. When summed, avoidable underallocation (i.e. underallocation that is not inevitable due to lower-than-needed resources) is found to be 4,175 units.

Overall underallocation is calculated by summing this red-shaded underallocation across all counties (17,324 units of resource). On a similar graph of the oracle distribution (not shown), the red shaded areas will represent unavoidable underallocation, as the oracle distribution is assumed to be ideal. Given a resource constraint of 90%, it follows that an oracle scheme will inevitably underallocate resources by 10%, and so unavoidable underallocation is known to be 10% of true need (13,149 units). Allocation score is then calculated as the difference between these two quantities (total underallocation minus unavoidable underallocation; 4,175) to estimate avoidable resource underallocation due to the forecast predictions.

Notably, overallocation (shown in blue) is not directly counted by AS, but due to resources being finite, a forecast with a large amount of overallocation will inevitably have a worse allocation score, due to incorrectly distributed resources.

In the visualization above, marked overallocation of resources to San Bernandino, Stanislaus, and Kern Counties can be identified, as well as underallocation to San Diego County and the counties from Santa Clara County to San Mateo County on the graph. The use of the allocation score framework and its associated visualizations has the added benefit of facilitating such qualitative assessments and can therefore assist in identifying targets for which a forecast's predictions are weakest. Review of these results can help to identify data artifacts or other threats to forecast accuracy. [**CA PEOPLE KNOW THE SOURCE OF THESE OVERPREDICTIONS? COULD POINT TO AS ILLUSTRATIVE**]

In practice, decision-makers may lack a definitive estimate of true resource availability. Furthermore, the calculation of AS from a single resource constraint may fail to capture resource distribution dynamics at higher or lower constraints. For this reason, the integrated allocation score is produced as an average of

a range of possible resource constraints, which can be weighted based on perceived probability of each constraint. For this example, an unweighted mean (uniform weighting) of 41 allocation scores from 70-110% of true need (92,044-144,642 units of resource) were calculated to produce an IAS of 12,284. As with AS, lower scores of IAS are indicative of better performing forecasts, and so this score may be compared to those of other forecasts for the same target date to assess model performance.

Methods

In this case study, forecasts of county-level incident COVID-19 cases during the omicron wave were evaluated for the state of California using IAS and MWIS.

Data Sources

Forecast data were sourced from the COVID-19 Forecast Hub, a central repository of forecasts and predictions submitted by international research groups.("Data - COVID 19 Forecast Hub," n.d.) Submissions are compiled in a standardized format for defined targets, locations, and time horizons, allowing for visualization, scoring, and analysis of forecasts. All forecast data are required to include predictions at 7 quantiles in the predictive distribution in order to be accepted to the Hub. (**covid19?**-)

Forecasts from the Hub's Zoltar repository [**CITE**] were extracted for a total of 13 target dates covering the duration of the omicron wave (12/18/2021 to 03/12/2022; MMWR weeks 2021-50 to 2022-12), at both the 1 and 3 week horizons. Target of interest was the county-level weekly incident COVID-19 case counts for all California counties with the exception of Los Angeles County (n=57), which was excluded due to its large population size (meaning that statewide IAS performance was expected to be dominated by performance in the county) as well as due to the existence of significant resource lines from the federal government [**CITE**]. Since most other counties in the state receive a majority of their resources from the State of California, Los Angeles County's allocation framework was deemed significantly different from the other counties as to justify its exclusion from the analysis. Through dropping this county, resource allocation could be conceived as coming from one centralized decision-maker, and COVID-19 case count could be used as a direct proxy for resource need.

Ground truth counts of county-level incident cases were retrieved through the JHU COVID-19 Data repository [**CITE**] via the covidHubUtils package [**CITE**]. For the calculation of population-based resource allocations, country-level population estimates for January 1st, 2021 were sourced from the State of California Department of Finance.("E-4 Population Estimates for Cities, Counties, and the State, 2021-2026 with 2020 Census Benchmark | California Department of Finance," n.d.).

Model Selection

Forecast submissions were deemed eligible for analysis if they included predictions for all 57 counties of interest on a given target date and horizon. Additionally, Forecast Hub ensembles (created via differing ensembling mechanisms from Hub forecast submissions) were dropped with the exception of the COVIDhub-4_week_ensemble.[CITE DOCUMENTATION WHAT MAKES IT SPECIAL] Additionally, the CU-nochange, CU-scenario_low, and CU-scenario_mid models were dropped, leaving only CU-select from among the compartmental forecasts submitted by Columbia University. According to the creators, the CU-select model was considered to be the most plausible scenario from among their submissions.(Pei and Shaman, n.d.) [CITE ZOLTAR]

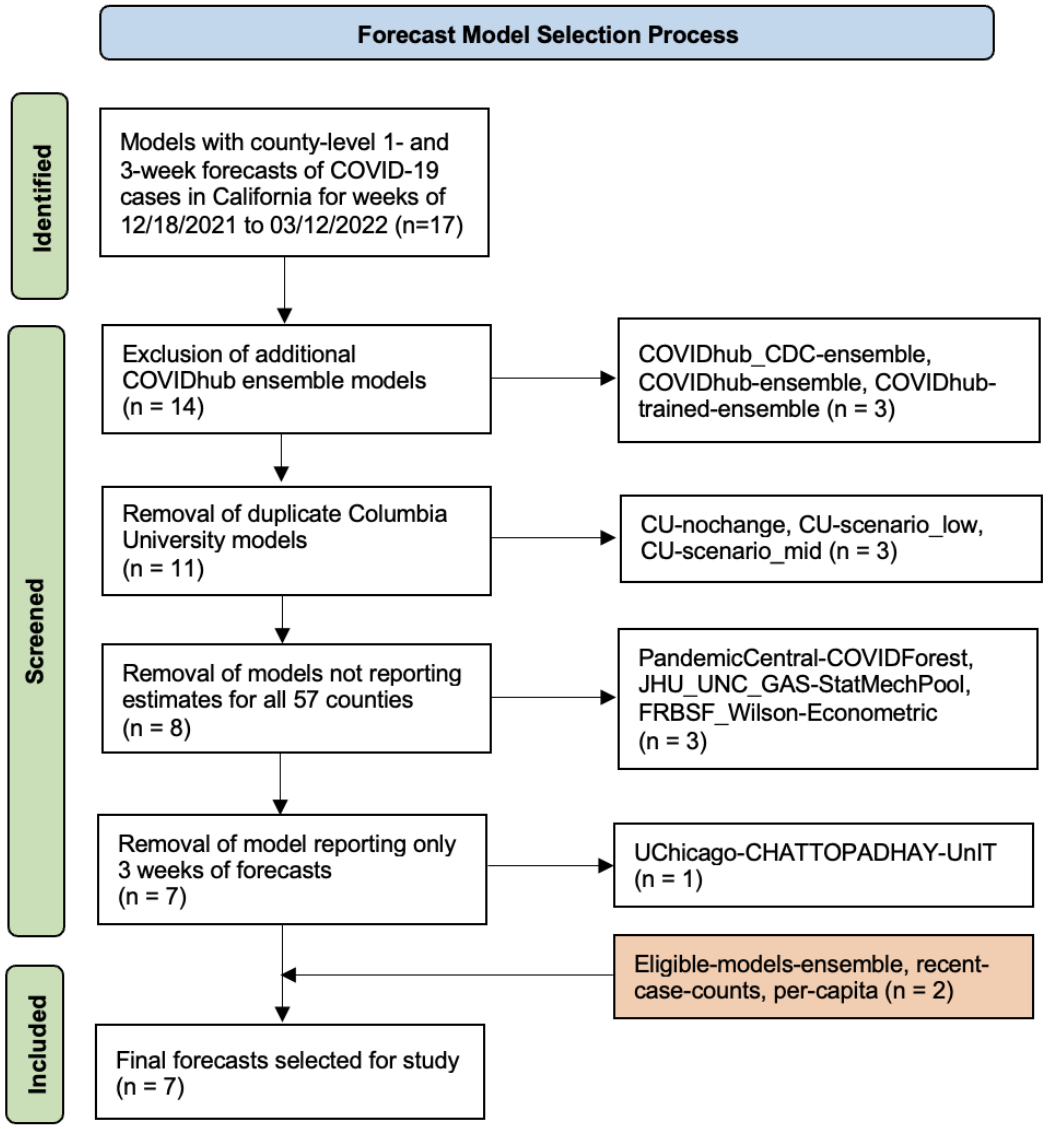


Figure 3: Selection process of models for inclusion in the allocation scoring analysis. Forecasts were eligible if they submitted predictions at the 1- and 3-week forecast horizons for all 57 California counties to the COVID-19 Forecast Hub for at least 4 weeks during the period of 12/18/2021 to 03/12/2022. Additional models from Columbia University were removed to include only their preferred forecast (CU-select). Ensembles from the Forecast Hub were also dropped, with the exception of the COVIDhub-4_week_ensemble model. Finally, additional forecasts/allocation schemes were produced for scoring: an ensemble forecast produced via uniform-weighted averaging of each date's non-Hub ensemble qualifying submissions (eligible-models-ensemble), resource allocation proportional to the most recent county case counts available at time of forecast (recent-case-counts), and resource allocation proportional to census population that ignores any epidemiological information (per-capita).

In addition to eligible individual forecast submissions, an ensemble forecast, named the "eligible-models-ensemble," was generated as uniform weighted average of all models with eligible submissions for each respective target end date and horizon, excepting the Hub's existing ensemble forecast (COVIDhub-4_week_ensemble). Quantile estimates for the eligible-models-ensemble were produced as an unweighted average of qualifying submissions for that quantile - for example, to produce a 0.90 ensemble estimate for a

given county, horizon, and target date, the mean was calculated from the 0.90 estimates from all eligible models submitting for that county, horizon, and date.

Finally, 2 benchmark allocation schemes were evaluated: a "per-capita" allocation scheme, in which the units of resource were allocated proportionally to estimated county population and ignoring epidemiological data, and a "recent-case-counts" scheme, n which the units of resource were allocated proportionally to the most recent case data for each country at the date of forecast (equivalent to the 0.500 quantile estimate of the COVIDhub-baseline model).

Statistical Analysis

Resource Distribution from Probabilistic Forecasts

In order to score the forecasts, optimized resource allocation schemes had to be produced from predictions reported in a quantile format. The method of resource allocation included in the `alloscore` package's `alloscore()` function was utilized for this process.(Gerding et al. 2024)

In this method, first, the set of quantiles submitted to the Hub is used to interpolate a cumulative distribution function (CDF) of expected case counts for each forecast submission. Next, using the calculated CDFs of the locations of interest and a set resource constraint, an iterative process is used to solve for a forecast-informed allocation of resource across the locations. At a given probability level, the predicted case counts from the CDFs are summed over all locations to determine the total predicted resource need at the probability level. The algorithm then iterates, increasing or decreasing the probability level across CDFs in parallel to solve for a probability level that approximates the total resource constraint. The resulting solution, expressed as a set of predicted case counts (i.e. predicted resource needs) across locations at the optimized probability level, are considered the forecast's allocation scheme for the target date and horizon. These allocation schemes are then scored using the allocation score procedure.

Integrated Allocation Score

For each respective target date and horizon, integrated allocation scores for forecasts were calculated using resource constraints ranging from 70% to 110% of ground truth case count (true need) at the respective target date (one allocation score for each percent between 70% and 110%, centered at 90% of true need, n = 41). A uniform-weighted integrated allocation score was used for most analyses, calculated as a mean of the allocation scores at each of the 41 resource constraints. To evaluate the impact of probability weighting on IAS, uniform-weighted scores were compared to a normal-weighted score, calculated using a normal distribution centered on the mean resource constraint (90% of true need for each forecast date). For each forecast date, standard deviation was calculated by assuming that the constraints at the 2.5th and 97.5th percentiles are 4 standard deviations apart (as dictated by a standard normal distribution) and thereby calculating standard deviation as the difference between these values divided by 4.

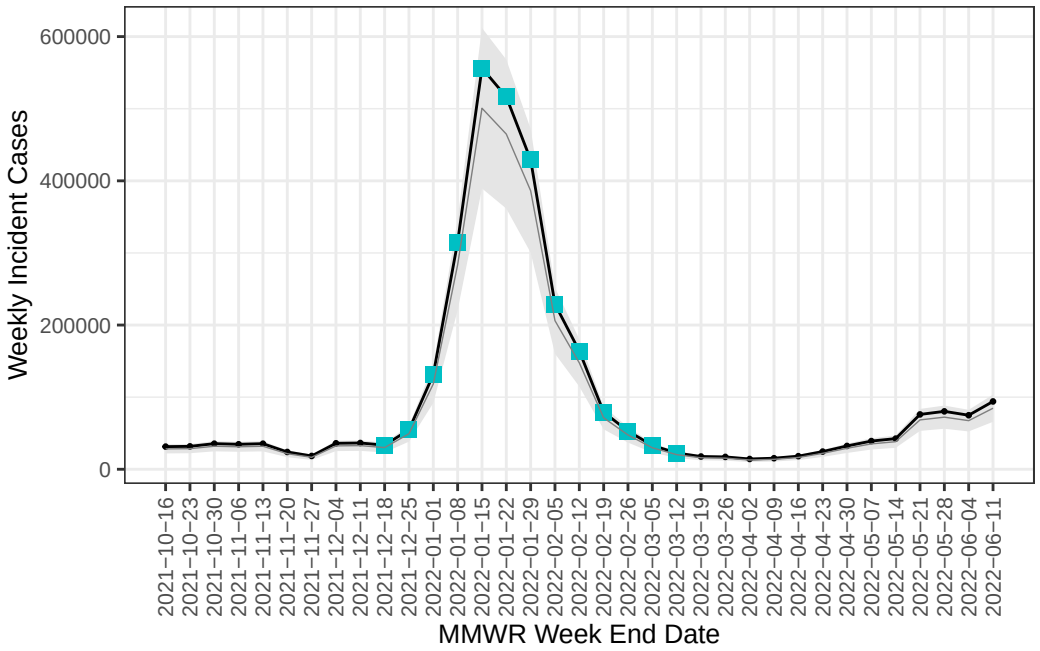


Figure 4: Omicron Wave in California: Allocation Score Resource Constraints. The grey line represents 90% of true need, while the shaded area bounds 70%-110% of true need. Allocation scores were determined at 41 different resource constraint levels between 70% and 110% and averaged using both uniform- and normal-weighting to calculate integrated allocation scores.

Comparisons Between IAS and Other Scores

In addition to uniform-weighted IAS and normal-weighted IAS, forecasts were evaluated via weighted interval score (WIS) for each target date, county, and horizon.(Bracher et al. 2021) Mean WIS (MWIS) was calculated as the unweighted average WIS score across all 57 counties for a given target date and horizon. Spearman's ρ was used to evaluate pairwise agreement in scores of models between scoring measures, and Kendall's τ was used to evaluate agreement in rankings.

Technical

All analyses were run in R version 4.6.1 using the `targets` package version 1.11.3 (Landau 2021). AS and IAS were calculated using the `alloscore` package (Gerding 2026) and WIS scores using the `scoringutils` package(Bosse et al., n.d.).

Results

Model Availability

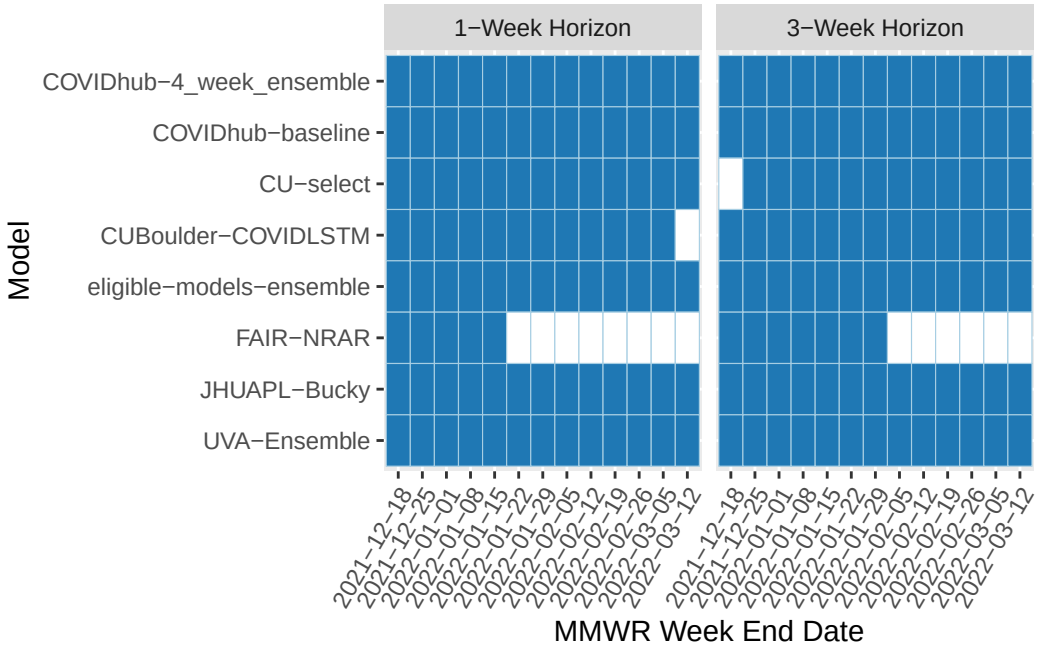


Figure 5: Complete forecast submissions by model and date. Forecast submissions were considered complete if they provided predictions for all 57 counties of interest. Hub validation procedures required that forecasts in quantile format provide predictions across 7 quantiles in order to be accepted.

At the 1 and 3 week forecast horizons, a total of 10 forecasts/schemes were evaluated, including the eligible-models-ensemble forecast and the per-capita and recent-case-counts schemes. Figure 4 outlines eligible forecast submissions (defined as submissions providing the required quantiles for all 57 counties of interest) by target date. Following model selection, all eligible submissions were used in the analysis. All eligible models for a given target and horizon were incorporated into the eligible-models-ensemble, with the exception of the COVIDhub-4_week_ensemble, which had been produced via ensembling methods. **[model order in viz? ADD PER CAP AND MOST RECENT DATA AS SCHEMES]**

Integrated Allocation Score

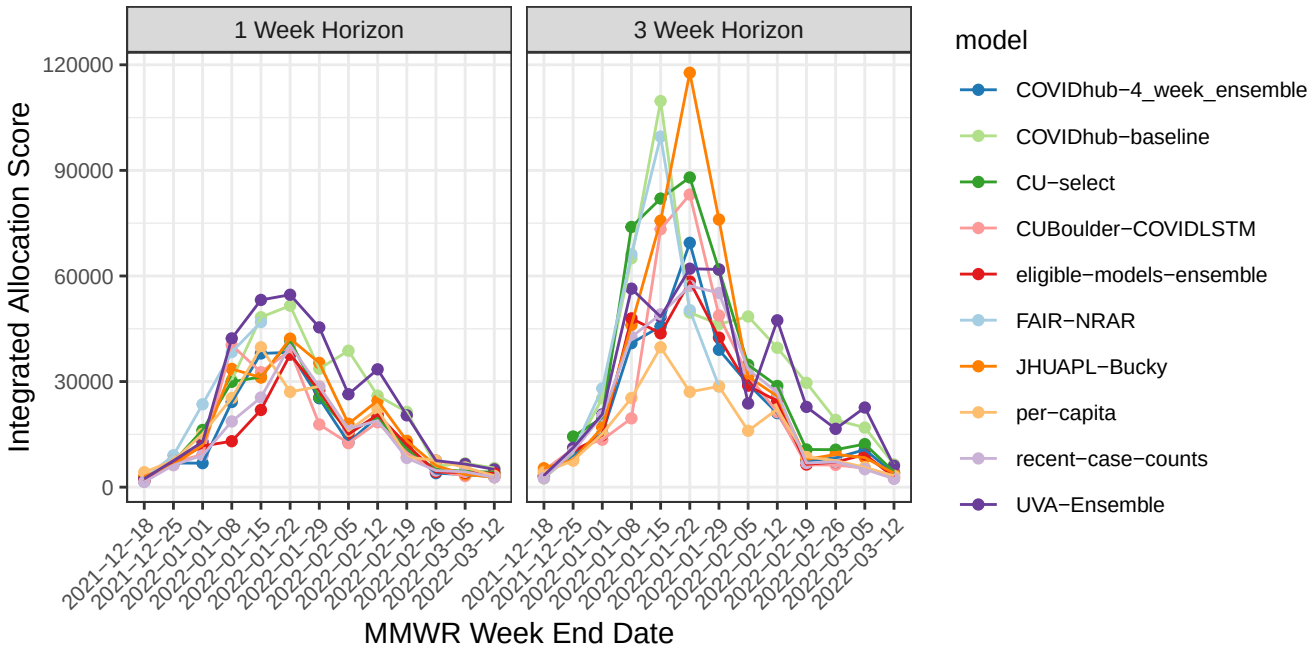


Figure 6: Integrated allocation scores over time, at the 1- and 3-week horizons, based on uniform-weighted averages of the 70% - 110% resource constraint levels. Scores at both horizons are seen to track roughly with the epidemic curve. The forecasts frequently fail to outperform the per-capita resource allocation scheme (shown in pale orange) at the 3-week horizon, particularly at the peak of the epidemic.

Uniform-weighted IAS results were found to vary over time for each model (Figure 5), but largely scaled with the epidemic curve. At the 3-week horizon, the IAS values were typically found to be higher than those at the 1-week horizon, reflecting poorer forecast performance. At both horizons, the scores of the forecasts as a whole were found to peak at the weeks of January 15th and 22nd and be lowest at the start and end of the omicron wave, in general alignment with the epidemic curve. Scores of individual forecasts, however, varied in their highest- and lowest-scoring weeks, as well as the shapes of their curves.

Being based on census populations instead of epidemiological data, the per-capita resource allocation scheme was used as a control. At the 1-week horizon, models scored better than per-capita more frequently than at the 3-week horizon (58.3% at the 1-week horizon, 21.8% at the 3-week horizon). At the 1-week horizon, 6 of the 9 submitting models (COVIDhub-4_week_ensemble, CUBoulder-COVIDLSTM, CU-select, eligible-models-ensemble, JHUAPL-Bucky, recent-case-counts) outscored per-capita in more than half of their predictions. In contrast, at the 3-week horizon, only 1 model, CUBoulder-COVIDLSTM, scored better than per-capita in more than half of its predictions. Of the models, 4 outperformed per-capita in more than half of their predictions over both horizons: COVIDhub-4_week_ensemble (53.9%), eligible-models-ensemble (57.7%), recent-case-counts (61.5%), and CUBoulder-COVIDLSTM (64.0%).

Of the forecast dates, the week of December 18th had the highest proportion of predictions (15 of 17 predictions, 88.2%) outperforming the per-capita allocation in IAS, followed by February 26th (11 of 16 predictions, 68.8%) and March 5th (8 of 16 predictions, 50%). In contrast, forecast predictions for the peak weeks of the epidemic (January 15th, 22nd, and 29th) outperformed per-capita a third of the time or less. Disparities between IAS values of the per-capita allocation scheme and the other models were particularly dramatic at the 3-week horizon during the peak weeks of the epidemic.

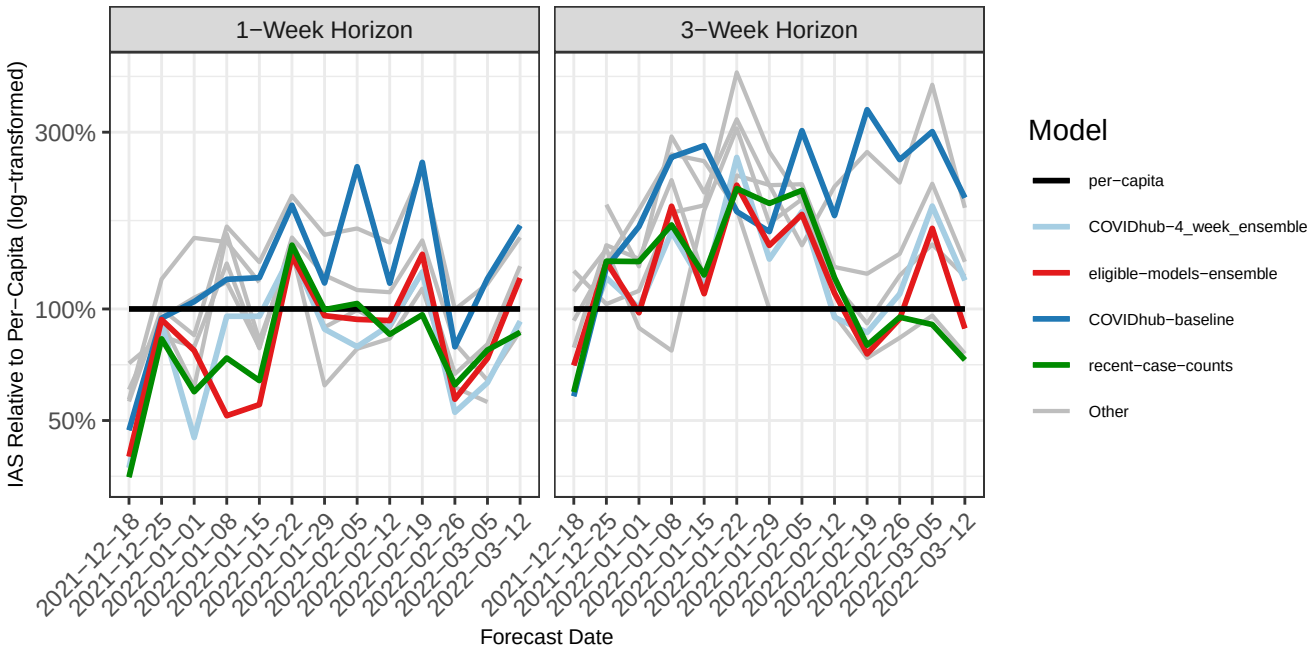


Figure 7: Uniform-weighted IAS for selected forecasts/allocation schemes as a percentage of the score of the per-capita allocation scheme, by target date. At the 1-week horizon, both the ensemble models and the recent-case-counts allocation scheme are found to outperform the per-capita early in the epidemic. At the 3-week horizon, the per-capita scheme is found to outperform the forecasts dramatically at almost all target dates. Noteably, IAS scores of the recent-case-counts allocation are found to track very closely with those of the ensemble models, while the COVIDhub-baseline model (which uses the same recent case counts as its point estimate, but includes a prediction interval), shows worse performance.

Figure 6 displays the IAS values as a percentage of the IAS of the control per-capita allocation, with select models highlighted (COVIDhub-4_week_ensemble, COVIDhub-baseline, eligible-models-ensemble, and recent-case-counts). In this visualization, lines under the bolded black line (representing 100% of the per-capita IAS) indicate better performance. Both ensemble models were found to frequently score lower than the per-capita allocation at the 1 week horizon, but not at the 3-week horizon. Scores were notably higher than the per-capita score on January 22nd (the peak of the epidemic) at the 3-week horizon, with all forecasts scoring at least double that of the per-capita allocation via IAS. The COVIDhub-baseline model was found to be outscored by the per-capita allocation and both ensembles for nearly all forecast dates, but disparities were particularly dramatic at the 3 week horizon. **[CUBoulder-COVIDLSTM]**

Comparisons Between IAS and Other Scores

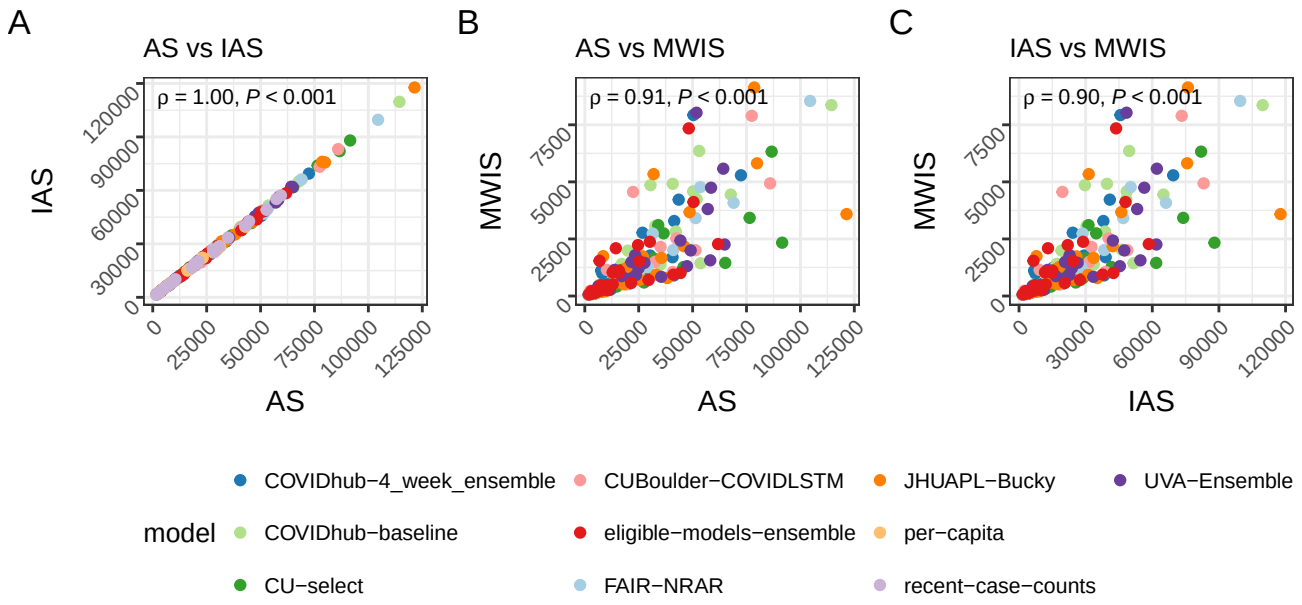


Figure 8: Scatter plots of pairwise comparisons between scoring measures: (A) AS vs IAS, (B) AS vs MWIS, and (C) IAS vs MWIS. In all 3 comparisons, and particularly in the comparison between AS and IAS, strong positive correlations were found via Spearman's ρ.

Comparisons between scoring metrics found strong monotonic correlations between the scoring metrics, with low scores in one metric predictive of low scores in the other two metrics. In Figure 7, scores are compared between pairs of metrics via scatter plot, with each dot indicating the scores for a single submission. AS and IAS were found to have a near-perfect correlation by Spearman's ρ ($\rho = 1.00$, $p < 0.001$), though both AS and IAS were also found to be strongly positively correlated with MWIS ($\rho = 0.91$ and 0.90, respectively). No single model was notable for its outliers or deviations from the trends seen on the plots.

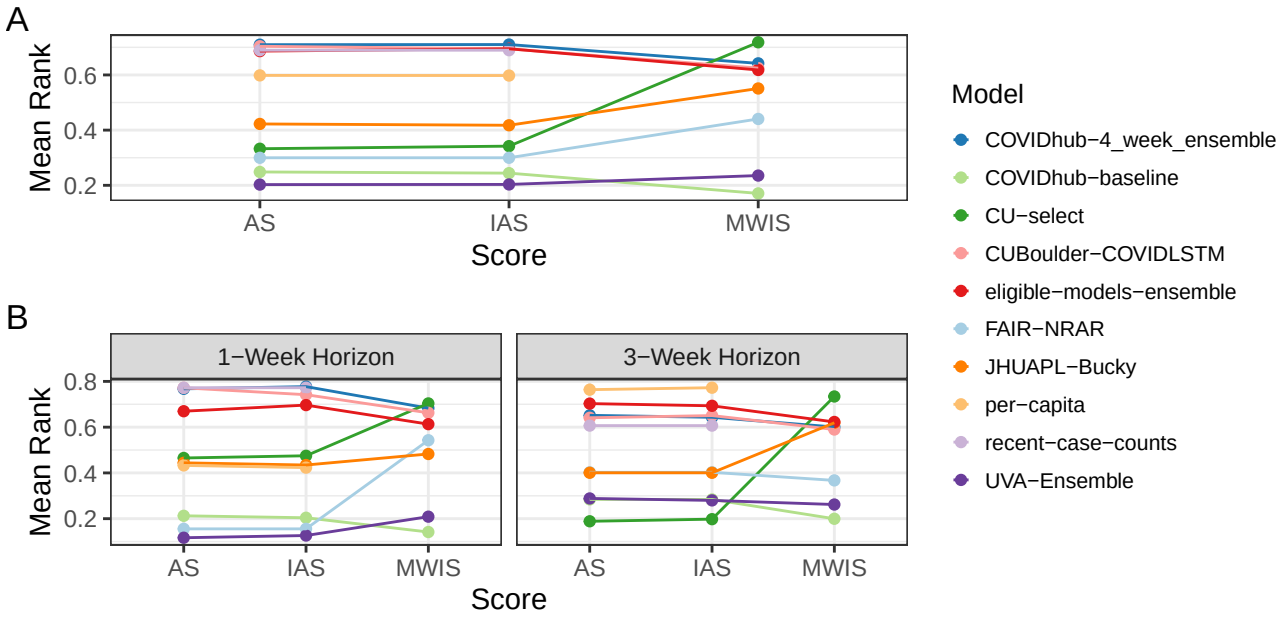
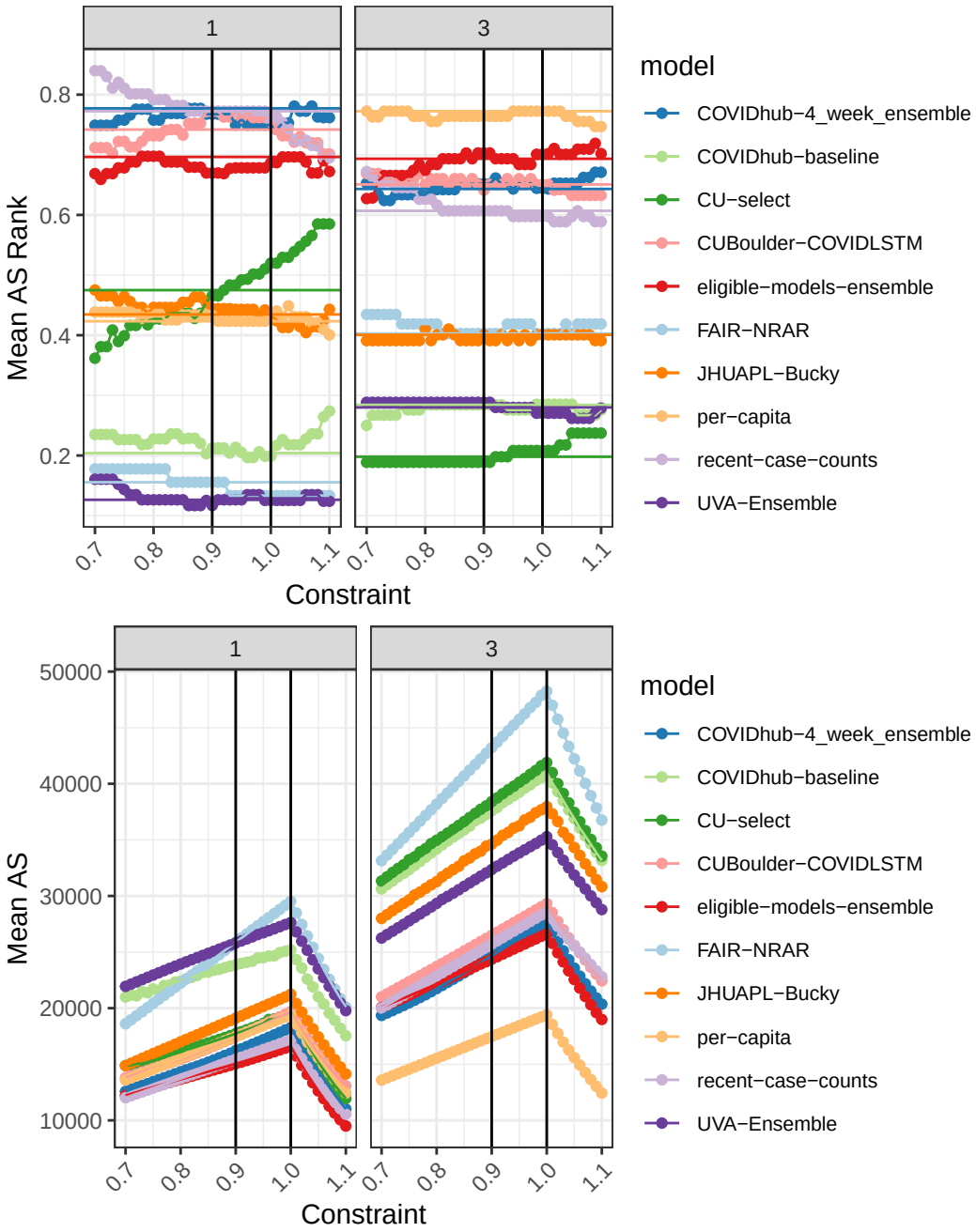


Figure 9: Mean rank (over dates and horizons (A) and over dates(B)) of models by scoring metric for AS, IAS, and MWIS, by model. Because the recent-case-counts and per-capita allocation schemes are generated without the use of prediction intervals, MWIS was not calculated. The ensemble models (COVIDhub-4_week_ensemble and eligible-models-ensemble) are among the best-ranking models across all 3 scoring metrics, alongside CUBoulder-COVIDLSTM. The per-capita allocation scheme was also found to rank highly in AS and IAS, though MWIS could not be calculated. Models showing high mean rank in one scoring method are typically found to be among the highest ranking in other metrics, though outliers exist, such as FAIR-NRAR at the 1-week horizon, and CU-select at the 3-week horizon.

A comparison of mean ranks of models between scoring metrics (AS at the 90% resource constraint, uniform-weighted AS, and MWIS) found that mean ranks were consistent between AS and IAS, but sometimes differed in MWIS. In Figure 8, mean relative ranks (0 = worst scoring model, 1 = best scoring model) for the 3 scoring metrics are visualized, first averaged across both forecast dates and horizons (A), and then faceted by horizon (B). High- or low-ranking in a score was often maintained across the other scores: for example, the COVIDhub-4_week_ensemble, eligible-models-ensemble, recent-case-counts, and CUBoulder-COVIDLSTM forecasts were found to be among the best-ranking models across all 3 scoring metrics, while COVIDhub-baseline and UVA-ensemble were consistently among the lowest in mean rank. However, some mean ranks were notable exceptions: CU-select, FAIR-NRAR and JHUAPL-Bucky were found to have low mean ranks in AS and IAS but higher ranks in MWIS, due to ranking highly in MWIS at the 1-week (CU-select and FAIR-NRAR) and 3-week (CU-select and JHUAPL-Bucky) horizons.



Uniform- vs. Normal-Weighting in IAS Calculation

Kendall's τ was used to evaluate agreement between model score rankings using IAS calculated via uniform weighting (mean) versus a normal distribution. Weights were centered on the mean resource constraint for each forecast date (90% of true need at that date) and standard deviation was determined from the 2.5th and 97.5th percentile values of the set of resource constraints used, assumed to be 4 standard deviations apart on a standard normal distribution. IAS values between the two weighting methods were found to be highly concordant in their model rankings, with τ values of 0.991 for IAS scores at the 1 week horizon, and 0.992 for scores at the 3 week horizon.

Discussion

In this analysis, we demonstrated the use the integrated allocation score to score forecasts of county-level incident COVID-19 case data during California's omicron wave. IAS values were found to scale approximately with the epidemic curve, with the highest scores for predictions of the wave's peak weeks, and scores at the 3-week forecast horizon were noticeably higher than those at the 1-week horizon, indicating poorer forecast performance at longer timescales. Comparison of forecast scores to controls found that forecasts frequently failed to outperform epidemiologically-naive resource allocation by population or by recent case data, suggesting areas for future forecast optimization. Finally, high-ranking models in IAS and AS generally were found to rank highly in MWIS and weighting method for IAS (uniform vs normal) was found to have little impact on model rankings.

Scores

Absolute AS and IAS scores largely aligned with expectations for the allocation score as introduced by Gerding et al.(Gerding et al. 2024). As seen in the original analysis, AS values were typically highest when the resource constraint was closest to the ground truth, peaking at the 100% constraint. [**INCLUDE IN SUPP?**] Additionally, absolute AS and IAS values scaled with the epidemic curve, as scoring for each forecast date was conducted using constraints proportional to the ground truth, and so misallocation could be expected to scale accordingly. Finally, forecast scores were found to be higher at the 3-week horizon than the 1-week horizon, in alignment with the general trend of better forecast performance at shorter time horizons. [**CITE**]

Rankings

The comparison of forecast IAS values to those of control allocation schemes allowed for the assessment of value-added by forecasts in informing resource allocation decisions. In Figure 10, resource allocation strategies are arranged in increasing order of information synthesis. In the per-capita scheme, resources are distributed proportionally to county census population, agnostic of current epidemiological trends, while in the recent-case-counts scheme, resources are distributed proportionally to the most recently reported case counts at the time of forecast. Forecasts and ensemble forecasts introduce additional complexity, generating allocation decisions through the anticipation of future trends and dynamics. To do so, they utilize past information, known population and environmental parameters, and (in the case of ensemble forecasts) other forecasts. This hierarchy of information allows forecasts to be assessed for whether their additional complexity contributes to better allocation decision-making, when compared to less analytically sophisticated approaches. Both control allocation methods are common approaches to resource allocation in public health practice, and more sophisticated methods should aim to improve allocation accuracy over these. [**CITE**]

In this analysis, forecasts frequently failed to outperform either the per-capita or recent-case-counts control when evaluated by IAS. The worst performance, when visualized as IAS relative to the IAS of the per-capita allocation, was found to occur with target dates at the peak of epidemic, when public health decisions can be considered most critical due to high resource burdens and morbidity. During the periods prior to the epidemic and during the decrease, relative IAS results were mixed. From this, we conclude that many of the county-level forecasts would not have improved allocation decisions during this critical time period, compared to the control methods. Notable exceptions were the two COVIDhub ensemble forecasts, COVIDhub-4_week_ensemble and eligible-models-ensemble, and CUBoulder-COVIDLSTM, which frequently scored similarly to or better than the controls, though much of their better performance occured during the weeks outside of the epidemic's peak.

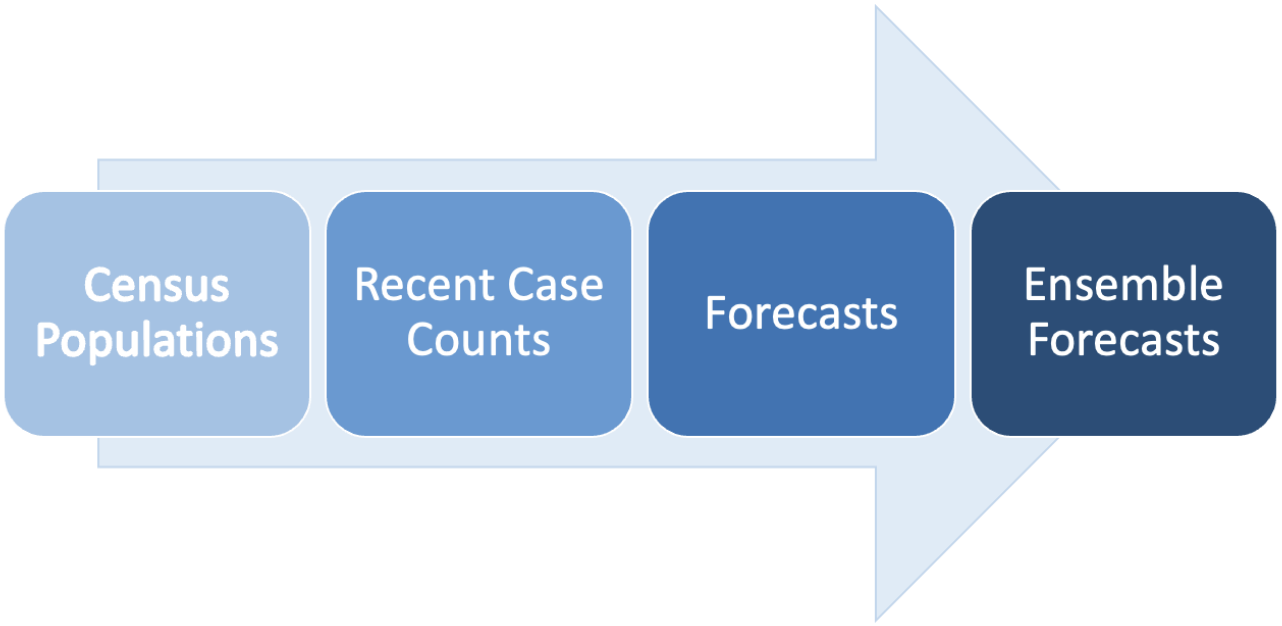


Figure 10: Hierarchy of increasing information complexity in epidemic response resource allocation strategies. Census population-based allocation ignores all epidemiological information and relies entirely on population sizes. Allocation based on recent case counts reflects epidemiological conditions, but may be subject to data backfill and future changes in dynamics across locations. Forecasts attempt to account for backfill and often utilize multiple data streams to make their predictions for each location. Ensemble forecasts compile submissions from multiple forecasting teams in order to produce their predictions. In this analysis, allocations via census population (per-capita) and recent case reports (recent-case-counts) were used as controls against which forecasts and ensemble forecasts were evaluated.

The comparison of mean relative rankings of models betwen evaluation methods found general agreement between methods, but some notable exceptions. AS, IAS by uniform weighting, and IAS by normal-weighting produced equivalent model rankings. When scored by MWIS, another commonly-used forecast evaluation metric, rankings also remained relatively consistent, though some models were found to rank more highly than they had with the allocation scoring methods. Taken together, this broad alignment in rankings indicates a general agreement between the methods and weighting strategies about "better" models, though metrics may be found to disagree occasionally, due to differences in which qualities are valued by their methodologies.

Limitations

This analysis contains several limitations. First and foremost, scoring by the integrated allocation score, while it attempts to ground forecasts in a decision-making context, still describes only abstract resources and distribution scenarios. Constraints used in this analysis were not intended to pertain to any specific apportionable resource, such as medicine or staffing, but rather were abstract units determined post-hoc from ground truth case counts and presented on the same unit scale. The translation of abstract resource units into real resources runs the risk of introducing feedback loops into the scoring method thought experiment For example, a forecast-informed distribution of PPE could impact disease transmission in the subsequent weeks and quickly render earlier forecasts invalid. Because of this, the scores cannot be translated easily into waste in units of funding, person-hours, or other resources. Further research would be necessary in order to bridge the gap between theoretical IAS and units of real world resources. However,

qualitative assessments via an allocation score-like framework, such as the use of visualizations like Figure 1 (see Introduction), may be useful in order to identify "weak points": locations to which forecasts and policies alike frequently under/overallocate. Such an analysis may be of interest to public health users, as the quality and granularity of forecasts of small-population subgroups (such as counties) has been identified as an area of concern and a potential barrier to forecast adoption.(Rao et al. 2025)

In addition, the distribution schemes scored in this analysis were assumed to come from perfect compliance to forecast predictions and their associated CDFs, which overlooks the complexity of real-world allocation schemes and the nuance with which forecasts are typically interpreted by officials. In practice, public health officials more frequently synthesize multiple information sources and models in order to make resource distribution decisions.[**CITE**] Furthermore, these decisions may be swayed by political and institutional priorities, emotional reactions, news cycles, or personal bias.[**CITE**] For this reason, as well as the previously-noted use of abstract resource units in the analysis, we hazard against use of the allocation score to interpret these forecast-based allocations as feasible policies.

Finally, the exclusion of Los Angeles County due to differences in supply lines prevents generalization of the results to forecasting for the state as a whole. Results must be evaluated as pertaining to forecasts made for the entire state minus Los Angeles. [**OPTIONAL: SUPPLEMENTAL MATERIAL WITH SAME ANALYSIS WITH LA COUNTY? COULD RUN BUT NEED TO NOTE VERY BLUNTLY THAT IT SHOULD BE TREATED WITH CAUTION**]

Application

In spite of these limitations, this analysis highlights the successful application of the IAS to an epidemiologic context in which forecasts may be expected to inform allocation-related decisions, as well as possible directions for future work. The frequent failure of the forecasts in this study to outscore a population based-control suggests that it may be a meaningful to add IAS to existing routine forecast evaluation and optimization workflows. [**REQUIRES AN EASIER TO RUN IAS, HOWEVER**] Further, through the use of the COVID-19 Forecast Hub or a similar platform, would be possible to evaluate and compare multiple forecasts via IAS and produce ensemble forecasts that weight based on recent IAS performance, similarly to how the COVIDhub-trained-ensemble weights its constituent forecasts by the inverse of MWIS. The performance of such IAS-optimized models remains a possible avenue for future forecast development.

Because IAS rewards forecasts that correctly capture relative resource needs across locations, this scoring rule may prove meaningful to public health decision-makers evaluating forecasts as information sources for multi-location resource distribution efforts. This method produces a measurement of avoidable, unaddressed resource need in a scenario where the forecast directly determines policy, rewarding models that capture the relative rankings of needs correctly across the locations of interest. The direct translation of this scoring to politically-relevant metrics, such as money or resource waste, as well as the generation of models that optimize for resource allocation effectiveness, remain areas for future research and development.

Conclusions

In this analysis, we demonstrate the use the integrated allocation score to score forecasts of county-level incident COVID-19 case data during California's omicron wave. Relative rankings of models by IAS were found to largely agree with those by MWIS, a frequently-used scoring rule. Additionally, comparison of mean-weighted IAS to scores by normal-weighting determined that weighting method made little difference in relative IAS rankings of models. The evaluation of the scores relative to a control allocation scheme agnostic of epidemiological trends (per-capita) allowed for measurement of the value-added in using the forecasts in resource distribution decisions, compared to solely population-based decision-making. Forecasts

frequently failed to beat this control when evaluated by IAS, particularly at the height of the epidemic. From this, we infer that, compared to population-based allocation, the county-level forecasts would not have improved allocation decisions during this critical time period in the public health response. During the upswing of the epidemic and last weeks of the wave, however, these relative scores were mixed, suggesting possible value-added by some models.

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